

Examples of using WMS to access Simena

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Introduction

The scope of this document is to determine the method of access to the information generated by Simena.

If you want to access a listing of available data, you should make a [GetCapabilities](#) request.

If you want to query the corrected temperature for a concrete observatory in Peru, for example, you should make a [GetFeature](#) request with the following values:

- QUERY_LAYERS=MTG_CORRECTED_TEMPERATURE (as taken from the GetCapabilities)
- Since the WMS protocol uses BBOX+WIDTH+HEIGHT+I+J, use WIDTH=101&HEIGHT=101&I=5&J=50 with a small BBOX to query the desired coordinate.
For example:
- Observatory coordinates are x=1.25343 y=40.435325
- BBOX centered on that point would be: x-0.000001,y+0.000001,x+0.000001,y-0.000001 -> BBOX=1.25342,40.435326,1.25344,40.435324

If you want to get a map, you should make a [GetMap](#) request. The meaning of the colours can be obtained by a [GetLegend](#) request.

WMS GetCapabilities example

Asking for the general information about the server available data. Request:

<https://agcommander.simena.red/wms/wms/?SERVICE=WMS&VERSION=1.3.0&REQUEST=GetCapabilities>

Response (an WMS Capabilities XML document):

```
<?xml version="1.0" encoding="UTF-8"?>
<WMS_Capabilities xmlns="http://www.opengis.net/wms" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://www.opengis.net/wms http://www.opengis.net/wms/wml.xsd">
  <Service>
    <Name>meteoGRID</Name>
    <Title>Weather forecast</Title>
    <Abstract></Abstract>
    <CRS>EPSG:4326</CRS>
    <CRS>EPSG:3857</CRS>
    <CRS>EPSG:25830</CRS>
    <EX_GeographicBoundingBox>
      <westBoundLongitude>-180</westBoundLongitude>
      <eastBoundLongitude>180</eastBoundLongitude>
      <southBoundLatitude>-90</southBoundLatitude>
      <northBoundLatitude>90</northBoundLatitude>
    </EX_GeographicBoundingBox>
    <BoundingBox CRS="CRS:84" minx="-178.04821777413082" miny="-87.15158463433839" maxx="177.6145019524" maxy="177.6145019524">
      <BoundingBox CRS="EPSG:4326" minx="-178.04821777413082" miny="-87.15158463433839" maxx="177.6145019524" maxy="177.6145019524">
        <BoundingBox CRS="EPSG:25830" minx="-3.6120028414793307E8" miny="-2.837919762899621E8" maxx="7.718819762899621E8" maxy="7.718819762899621E8">
          <BoundingBox CRS="XEPSG:4326" minx="-87.15158463433839" miny="-178.04821777413082" maxx="87.6619262" maxy="87.6619262">
            <Layer queryable="0" opaque="0" cascaded="0">
              <Name>METEGRID_REGIONS</Name>
              <Title>Regional boundaries</Title>
              <Abstract>None</Abstract>
              <Style>
                <Name>default</Name>
                <Title>default</Title>
                <LegendURL width="233" height="625">
                  <Format>image/png</Format>
                  <OnlineResource xmlns:xlink="http://www.w3.org/1999/xlink" xlink:type="simple" xlink:href="http://www.w3.org/1999/xlink" />
                </LegendURL>
              </Style>
            </Layer>
            <Layer queryable="1" opaque="0" cascaded="0">
              <Name>HARMONIE_ARPEGE_ECMWF_TMAX</Name>
              <Title>Temperature</Title>
              <Abstract><b>Origen de la información</b>: Modelos Harmonie-Arpege-ECMWF. Pasada de las 0:00 UTC<br><b>Resolución temporal</b>: Diaria<br><b>Horizonte temporal</b>: 8 días</Abstract>
              <Style>
                <Name>default</Name>
                <Title>default</Title>
                <LegendURL width="233" height="625">
                  <Format>image/png</Format>
                  <OnlineResource xmlns:xlink="http://www.w3.org/1999/xlink" xlink:type="simple" xlink:href="http://www.w3.org/1999/xlink" />
                </LegendURL>
              </Style>
            </Layer>
          </BoundingBox>
        </BoundingBox>
      </BoundingBox>
    </EX_GeographicBoundingBox>
  </Service>
  <Capability>
    <Request>
      <Format>XML</Format>
      <Format>INIMAGE</Format>
      <Format>BLANK</Format>
    </Request>
    <Exception>
      <Format>XML</Format>
      <Format>INIMAGE</Format>
      <Format>BLANK</Format>
    </Exception>
  </Capability>
</WMS_Capabilities>
```

WMS GetFeature example

Asking for the evolution of a variable in a certain coordinate. As per WMS standard, the

requested point is relative (coordinates I, J of a WIDTH x HEIGHT) to a bounding box. A

TOLERANCE parameter is added in order to enclose near point features. Request:

https://agcommander.simena.red/wms/wms/?SERVICE=WMS&VERSION=1.3.0&REQUEST=GetFeatureInfo&QUERY_LAYERS=ARPEGE_TEMPERATURE&LAYERS=ARPEGE_TEMPERATURE&STYLES=&TIME=2023-01-19T12%3A00%3A00Z&INFO_FORMAT=application%2Fjson&TOLERANCE=3&HORIZON=1000&I=405&J=60&WIDTH=512&HEIGHT=512&CRS=EPSG%3A3857&BBOX=-1252344.271424327%2C3757032.814272983%2C6.984919309616089e-10%2C5009377.085697311

Legible values:

SERVICE: WMS

VERSION: 1.3.0

REQUEST: GetFeatureInfo

QUERY_LAYERS: ARPEGE_TEMPERATURE

LAYERS: ARPEGE_TEMPERATURE

STYLES:

TIME: 2023-01-19T12:00:00Z

INFO_FORMAT: application/json

TOLERANCE: 3

HORIZON: __all__

I: 405

J: 60

WIDTH: 512

HEIGHT: 512

CRS: EPSG:3857

BBOX: -1252344.271424327,3757032.814272983,6.984919309616089e-10,5009377.085697311

In this case, all temporal horizons from the calculation run in which said TIME is contained are retrieved.

Response (a GeoJSON document):

```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "FeatureCollection",
      "features": [
        {
          "type": "Feature",
          "name": "ARPEGE_TEMPERATURE",
          "geometry": null,
          "properties": {
            "base_time": "2023-01-19T00:00:00Z",
            "value": -0.04648923873901367,
            "HOR_ARPEGE": 0,
            "time": "2023-01-19T00:00:00Z"
          }
        },
        {
          "type": "Feature",
          "name": "ARPEGE_TEMPERATURE",
          "geometry": null,
          "properties": {
```

```

        "base_time": "2023-01-19T00:00:00Z",
        "value": -0.24009323120117188,
        "HOR_ARPEGE": 1,
        "time": "2023-01-19T01:00:00Z"
    },
    ...
],
    "layer_name": "ARPEGE_TEMPERATURE",
    "layer_title": "ARPEGE_TEMPERATURE",
    "has_time": true
}
]
}

```

When reading point features (as opposed to raster) the poi can be specified by id with the poi_id optional parameter.

WMS GetMap example

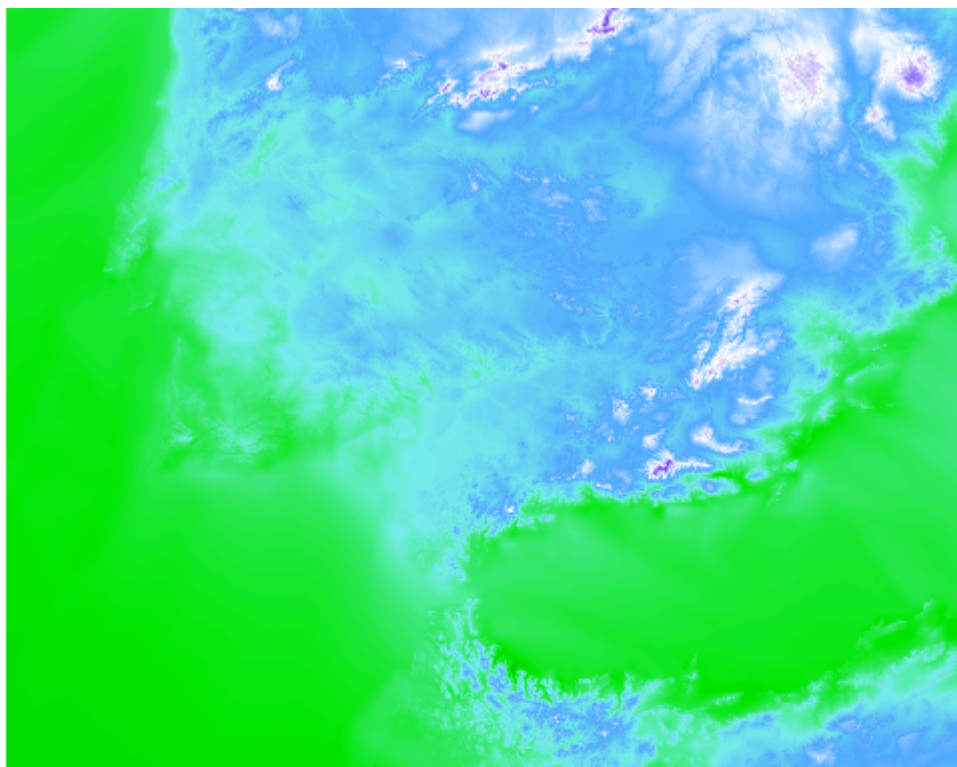
Asking for the coloured map of a certain Bounding Box. Request:

https://agcommander.simena.red/wms/wms/?SERVICE=WMS&VERSION=1.3.0&REQUEST=GetMap&FORMAT=image%2Fpng&TRANSPARENT=true&LAYERS=ARPEGE_TEMPERATURE&STYLES=&TIME=2023-01-19T12%3A00Z&WIDTH=512&HEIGHT=512&CRS=EPSG%3A3857&BBOX=-1252344.271424327%2C3757032.814272983%2C6.984919309616089e-10%2C5009377.085697311

Legible values:

SERVICE: WMS
VERSION: 1.3.0
REQUEST: GetMap
FORMAT: image/png
TRANSPARENT: true
LAYERS: ARPEGE_TEMPERATURE
STYLES:
TIME: 2023-01-19T12:00:00Z
WIDTH: 512
HEIGHT: 512
CRS: EPSG:3857
BBOX: -1252344.271424327,3757032.814272983,6.984919309616089e-10,5009377.085697311

Response (an image):



WMS GetLegend example

Asking for the legend of said map. Request:

https://agcommander.simena.red/wms/wms/?SERVICE=WMS&VERSION=1.3.0&REQUEST=GetLegendGraphic&FORMAT=image%2Fpng&LAYER=ARPEGE_TEMPERATURE&width=150&TRANSPARENT=TRUE

SERVICE: WMS

VERSION: 1.3.0

REQUEST: GetLegendGraphic

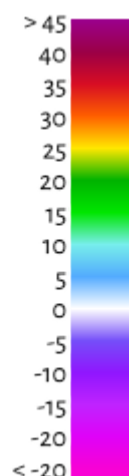
FORMAT: image/png

LAYER: ARPEGE_TEMPERATURE

width: 150

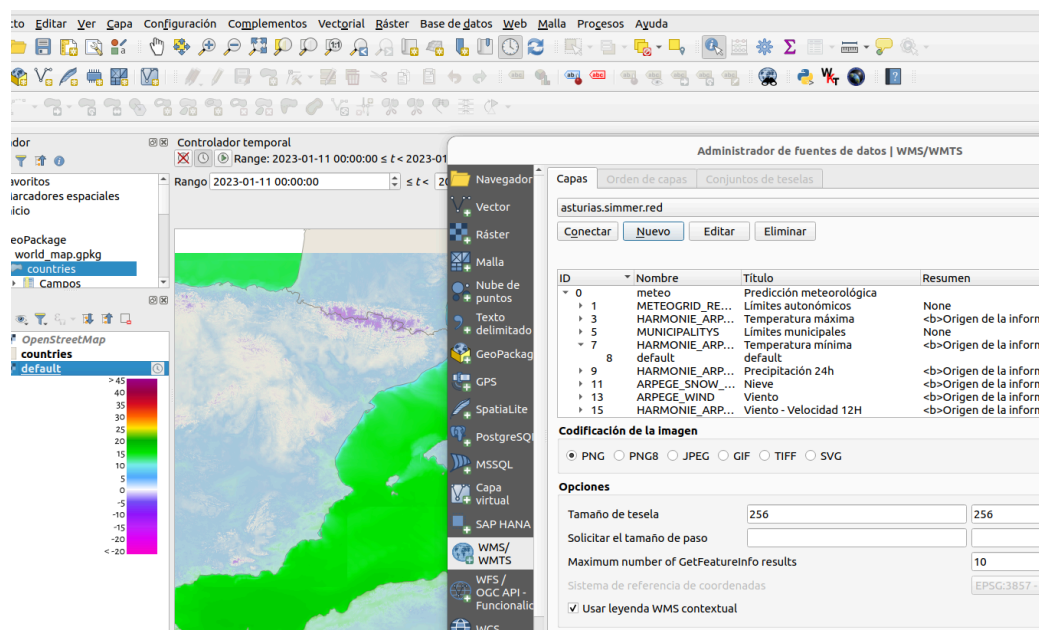
TRANSPARENT: TRUE

Response (an image):



Use in Desktop Client

The WMS Service can be used from desktop clients such as QGIS, ESRI, etc.



References

<https://www.ogc.org/standards/wms>

https://en.wikipedia.org/wiki/Web_Map_Service